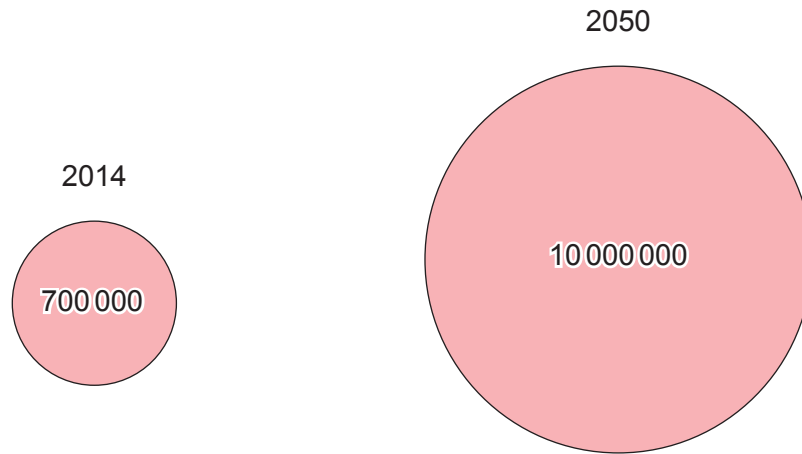


9. The term superbug is used to describe disease-causing (pathogenic) bacteria that are no longer killed by certain antibiotics. These bacteria have developed antibiotic resistance.

Image 9.1 shows the actual deaths from superbugs in the world in 2014 and the predicted deaths in 2050.

Image 9.1



[Source: United Nations]

- (a) (i) Use the equation below to calculate the percentage increase in predicted deaths from superbugs in the world between 2014 and 2050.
Give your answer to **the nearest whole number**. [3]

$$\frac{\text{number in 2050} - \text{number in 2014}}{\text{number in 2014}} \times 100$$

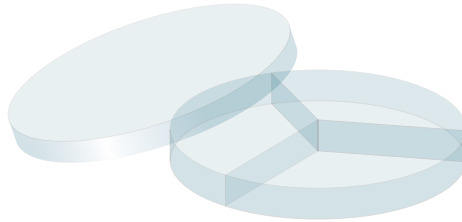
Percentage increase in deaths =

- (ii) State the main reason for the increase of antibiotic resistance in some pathogenic bacteria. [1]

.....



- (b) Health scientists were experimenting on the effect of the antibiotic methicillin on three different species of pathogenic bacteria. They used a triple compartment Petri dish in their experiment.

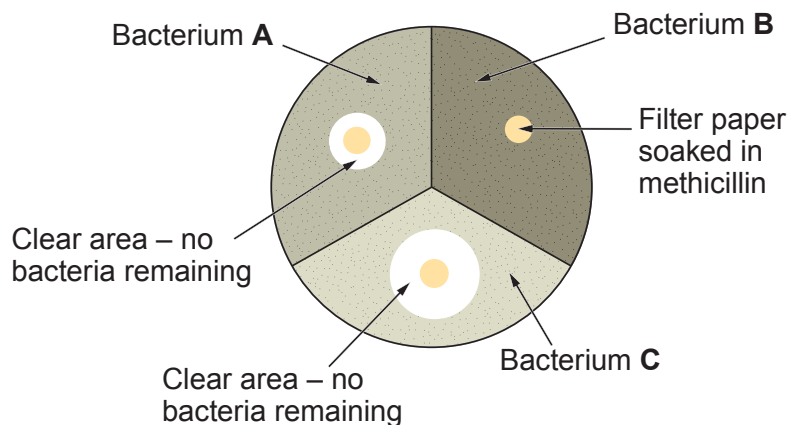


The method they used is shown below:

- Each compartment was filled with a nutrient jelly (agar), which allows bacteria to grow.
- Bacteria species **A**, **B** and **C** were each placed in a different compartment.
- A disc of filter paper containing the antibiotic methicillin was placed in the centre of each compartment.
- The Petri dish lid was used to close the dish. The dish was then placed in an incubator at 37 °C.
- After two days the Petri dish was examined.

Image 9.2 shows the surface view of the Petri dish after 2 days. The shaded areas show where bacteria were living. The clear areas show where there were no bacteria remaining.

Image 9.2



- (i) Use **Image 9.2** to identify which bacterium is the superbug MRSA.
(Circle) your answer.

[1]

A

B

C

